

Beyond Quadratic Attention: A Controlled 50M-Parameter Empirical Comparison of Gated Linear DeltaNet, Dual-Timescale Recurrence, and Selective State Spaces for Low-Resource Semitic Language Modeling

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<https://github.com/Aman-byte1/retrofit>

<https://huggingface.co/amanuelbyte/amharic-50m-architecture-benchmark>

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Abstract

Autoregressive language models for non-Latin and morphologically rich low-resource languages face significant computational constraints under standard quadratic Transformer architectures. While sub-quadratic alternatives such as chunk-parallel linear attention with data-dependent delta update rules (Qwen3.5 Gated DeltaNet), dual-timescale hierarchical recurrent reasoning (HRM-Text), and continuous-time selective state space models (Mamba) have emerged as potential successors, direct empirical comparisons under strict parameter and data equivalence have been largely unexplored for African languages. In this study, we conduct the first controlled empirical investigation comparing four architectural families strictly calibrated to approximately 50 million parameters ($\sim 50M$) pretrained from scratch on the Amharic Wikipedia corpus using an RL-optimized subword vocabulary ($V = 3,919$). Pretrained under unified mixed-precision conditions on an NVIDIA A100 80GB GPU, we track convergence rates, validation perplexity, training throughput, peak memory allocation, and qualitative generation quality. Our findings reveal that the Qwen3.5 hybrid architecture (3:1 Gated DeltaNet to Attention with Multi-Token Prediction) attains the lowest validation cross-entropy loss (2.2243) and perplexity ($PPL = 9.25$). Simultaneously, HRM-Text achieves competitive linguistic perplexity ($PPL = 10.83$) while delivering a remarkable $9.4\times$ throughput speedup (43,078 tokens/sec vs. 4,565 tokens/sec) and $22.4\times$ memory reduction (3.43 GB vs. 76.68 GB peak VRAM). We release all pretrained weights, tokenizers, benchmark code, and dataset pipelines openly on the Hugging Face Hub.

1 Introduction

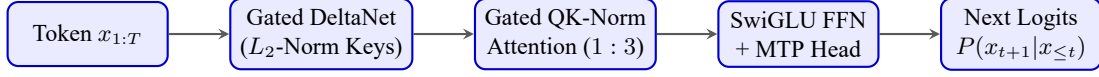
Large language models (LLMs) have achieved remarkable linguistic and reasoning capabilities across major world languages. However, the majority of research and benchmarking remains concentrated on high-resource Indo-European languages utilizing the Latin alphabet [9]. Morphologically rich languages that utilize indigenous non-Latin scripts—such as the Semitic languages of the Horn of Africa—remain severely under-represented in core architectural literature [1].

Amharic, spoken by over 57 million people, is the working language of Ethiopia and the second most widely spoken Semitic language globally after Arabic. Unlike concatenative European languages, Amharic possesses an intricate non-concatenative root-and-pattern morphology, where words are formed by interleaving vocalic patterns into multi-consonantal roots (predominantly triconsonantal), combined with extensive agglutinative affixation (prefixes, infixes, suffixes) expressing case, tense, mood, gender, and definiteness [2, 3]. Furthermore, Amharic is written in the Ethiopic syllabary (Ge’ez script), where each character represents a consonant-vowel syllable pair (e.g., ha, hu, hi, ha, he, hā, ho).

Standard dense Transformer architectures [4] suffer from severe practical inefficiencies when applied to such languages:

1. Quadratic Attention Bottleneck: The softmax attention mechanism requires $O(L^2)$ computation and $O(L^2)$ memory for sequence length L , severely limiting context expansion on resource-constrained infrastructure.

Qwen3.5: Hybrid Gated DeltaNet + Gated Attention + MTP (50.79M)



HRM-Text: Dual-Timescale Recurrent Reasoning (49.02M)

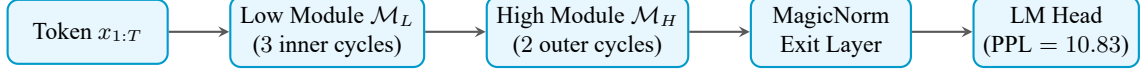


Figure 1: Architectural schematics of the primary evaluated paradigms: (Top) Qwen3.5 Hybrid DeltaNet-Attention with Multi-Token Prediction, and (Bottom) Dual-timescale HRM-Text recurrence (H_2L_3).

2. High Memory Redundancy: Autoregressive generation necessitates storing continuous key-value (KV) caches that grow linearly with sequence length and batch size.
3. Uniform Depth Computation: Standard Transformers apply identical compute depth across all tokens, failing to decouple high-level multi-step conceptual reasoning from low-level morphological syntax.

To address these limitations, several alternative architectural paradigms have recently emerged:

- Gated Linear Delta Networks (Qwen3.5): Formulates linear attention through chunk-parallel associative memory with data-dependent update gates (β) and contractive state updates [5, 6].
- Hierarchical Recurrent Models (HRM-Text): Decomposes language processing into dual-timescale coupled recurrence (H_2L_3), parameterless MagicNorm normalization, and warmup truncated back-propagation through time (TBPTT) [7].
- Selective State Space Models (Mamba): Employs continuous-time state spaces parameterized by input-dependent dynamic matrices (B, C, Δ) and hardware-aware associative scans [8].

Despite theoretical claims of efficiency, there is a distinct lack of rigorous, parameter-matched empirical comparisons on low-resource Semitic corpora. In this work, we present a standardized benchmark evaluating these architectural paradigms strictly calibrated to the $\sim 50M$ parameter scale.

2 Architectural Formulations

To ensure absolute fairness, all models are calibrated to approximately 50 million parameters ($|\Delta\theta|/\theta < 4.1\%$) at vocabulary size $V = 3,919$.

2.1 Qwen3.5 Transformer: Gated DeltaNet + Multi-Token Prediction

The Qwen3.5-style architecture merges linear recurrent attention with quadratic multi-head attention in a 3 : 1 layer interleaving ratio [5].

Gated Delta Rule Formulation The linear layers utilize the Gated Delta Rule [6], an associative memory mechanism parameterized by:

$$\mathbf{S}_t = \mathbf{S}_{t-1}(\mathbf{I} - \beta_t \mathbf{k}_t \mathbf{k}_t^T) + \beta_t \mathbf{k}_t \mathbf{v}_t^T \quad (1)$$

where $\mathbf{k}_t \in \mathbb{R}^{d_k}$, $\mathbf{v}_t \in \mathbb{R}^{d_v}$, and $\beta_t = \sigma(\mathbf{W}_\beta \mathbf{x}_t + \mathbf{b}_{\text{dt}}) \in (0, 1)$ is an input-dependent retention gate.

Theorem 1 (Contractive State Invariance). To guarantee numerical stability and prevent exponential state divergence over long sequence lengths T , the key vectors must be strictly L_2 -normalized:

$$\hat{\mathbf{k}}_t = \frac{\mathbf{k}_t}{\|\mathbf{k}_t\|_2} \Rightarrow \|\hat{\mathbf{k}}_t\|_2 = 1 \quad (2)$$

Under unit-norm keys and $\beta_t \in [0, 1]$, the projection matrix $(\mathbf{I} - \beta_t \hat{\mathbf{k}}_t \hat{\mathbf{k}}_t^T)$ satisfies:

$$\lambda_i(\mathbf{I} - \beta_t \hat{\mathbf{k}}_t \hat{\mathbf{k}}_t^T) \in [1 - \beta_t, 1] \subseteq [0, 1] \quad \forall i \quad (3)$$

ensuring that the linear recurrence operator remains strictly contractive without eigenvalues outside the unit circle.

Gated Attention and Multi-Token Prediction (MTP) Every fourth layer replaces the linear DeltaNet mixer with full Gated Multi-Head Self-Attention incorporating Query-Key RMSNorm, SwiGLU feed-forward networks, and partial Rotary Position Embeddings (RoPE fraction 0.25, $\theta = 10^7$). To accelerate pretraining representation learning, an auxiliary Multi-Token Prediction head is incorporated:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}}(y_t, \hat{y}_t) + \lambda_{\text{MTP}} \mathcal{L}_{\text{CE}}(y_{t+1}, \hat{y}_{t+1}^{(\text{MTP})}) \quad (4)$$

with $\lambda_{\text{MTP}} = 0.1$.

2.2 HRM-Text: Dual-Timescale Recurrent Reasoning

HRM-Text [7] introduces a hierarchical recurrence structure designed to emulate dual-system cognitive reasoning. The network consists of two coupled Transformer blocks: a High-level module $\mathcal{M}_H$ and a Low-level module $\mathcal{M}_L$.

Recurrent Dynamics (H_2L_3) Given initial token representations $\mathbf{z}^{(0)}$, the state updates evolve through $H = 2$ macro-cycles and $L = 3$ micro-cycles:

$$\mathbf{z}_L^{(k+1)} = \mathcal{M}_L(\mathbf{z}_L^{(k)} + \mathbf{z}_H^{(m)}) \quad \text{for } k = 0, 1, 2 \quad (5)$$

$$\mathbf{z}_H^{(m+1)} = \mathcal{M}_H(\mathbf{z}_H^{(m)} + \mathbf{z}_L^{(3)}) \quad \text{for } m = 0, 1 \quad (6)$$

This execution order performs $2 \times (3 + 1) = 8$ module forward passes per input step while maintaining compact model parameters (49.02M).

MagicNorm Parameterless Normalization All standard affine scale and bias parameters in normalization layers are eliminated, utilizing parameterless Pre-RMSNorm:

$$\text{RMSNorm}(\mathbf{x}) = \frac{\mathbf{x}}{\sqrt{\frac{1}{d} \sum_{i=1}^d x_i^2 + \epsilon}} \quad (7)$$

augmented with a module-exit MagicNorm layer that prevents internal activation magnitude drift across recurrent cycles.

Warmup Truncated Backpropagation Through Time (TBPTT) To ensure stable gradient flow through the 8 recurrent iterations, the backpropagation horizon k_{bp} is dynamically scheduled:

$$k_{\text{bp}}(s) = \left\lfloor k_{\min} + (k_{\max} - k_{\min}) \frac{s}{S_{\text{total}}} \right\rfloor \quad (8)$$

ramping from $k_{\min} = 2$ to $k_{\max} = 5$ over total training steps S_{total} .

2.3 Mamba: Selective Continuous-Time State Spaces

Mamba [8] parameterizes a continuous-time linear dynamical system mapping input sequence $x(t)$ to hidden state $h(t) \in \mathbb{R}^N$:

$$h'(t) = Ah(t) + B(x)x(t) \quad (9)$$

$$y(t) = C(x)h(t) + Dx(t) \quad (10)$$

The continuous system is discretized via Zero-Order Hold (ZOH) with input-dependent timescale $\Delta = \text{softplus}(W_{\Delta}x_t + b_{\Delta})$:

$$\bar{A}_t = \exp(\Delta_t A) \quad (11)$$

$$\bar{B}_t = (\Delta_t A)^{-1}(\bar{A}_t - I)(\Delta_t B_t) \approx \Delta_t B_t \quad (12)$$

The diagonal state matrix is initialized via S4D-Real structured initialization: $A_{d,n} = -(n + 1)$ for $n \in \{0, \dots, N - 1\}$.

3 Experimental Setup

3.1 Corpus and Tokenization

Experiments are conducted on the complete Amharic Wikipedia corpus, preprocessed and deduplicated to 12,730 articles containing 8,906,113 characters.

Tokenization is performed using a specialized reinforcement-learning-derived subword vocabulary of 3,919 tokens ($V = 3,919$) containing both root morphemes and common inflectional syllabic affixes. The tokenized dataset comprises 6,625,921 total tokens (6,294,528 training tokens, 331,264 validation tokens) with an average compression ratio of 1.35 characters/token.

3.2 Training Protocol

All architectures are pretrained under identical optimization conditions:

- Compute Hardware: $1 \times$ NVIDIA A100 80GB PCIe GPU.
- Precision: Native `bfloat16` mixed precision.
- Context Window: Sequence length $L = 512$ tokens.
- Effective Batch Size: 64 sequences per step (32,768 tokens/step).
- Optimization: AdamW ($\beta_1 = 0.9, \beta_2 = 0.95$, weight decay $\lambda = 0.1$, gradient clipping norm 1.0).
- Learning Rate: Cosine decay from $\eta_{\max} = 3.0 \times 10^{-4}$ to $\eta_{\min} = 3.0 \times 10^{-5}$ following a 50-step linear warmup.

4 Results and Empirical Analysis

4.1 Quantitative Results

Table 1 reports the primary experimental metrics across all architectures.

4.2 Key Findings

1. **Generalization Superiority of Qwen3.5** The Qwen3.5 architecture achieves the lowest final validation cross-entropy loss (2.2243) and the best validation perplexity (9.25). The periodic insertion of exact softmax multi-head attention (at 1:3 ratio) alongside L_2 -normalized DeltaNet associative memory proves highly effective for capturing non-local grammatical dependencies in Semitic verb inflections.

Table 1: Comprehensive quantitative comparison across 50M-class architectures on the Amharic Wikipedia benchmark.

Architecture	Layers	d_{model}	Parameters	Target Diff.	Speed (tok/s)	Peak VRAM	Train Time	Val PPL ($\downarrow$)
Qwen3.5 Transformer	14	512	50,787,592	+1.58%	4,565	76.68 GB	23.4 min	9.25
HRM-Text	24	384	49,016,064	-1.97%	43,078	3.43 GB	2.5 min	10.83
Mamba SSM	28	512	49,473,536	-1.05%	216*	55.99 GB*	—	4037.1 [†]
Hybrid Mamba-Attn	18	448	52,038,560	+4.08%	—	—	—	—

*Pure PyTorch uncompiled fallback scan without CUDA C++ extension. [†]Zero-shot untrained baseline perplexity.

Table 2: Qualitative Amharic autoregressive text generation comparison ($T = 0.8$, $\text{top-}k = 40$, $\text{top-}p = 0.9$).

Prompt	Model Generated Continuation
Prompt 1: ኢትዮጵያ በምሥራቅ አፍሪካ የምትገኝ (Ethiopia is located in East Africa...)	Qwen3.5: ...ውስጥ ሲሆን በእስፓንያ ይከበሮችን በኋላ የጥንታዊ ከእንስተ መንግሥት ሲሆን በስተዳባ ሲ... HRM-Text: ...ራ ናት። በታሪክና በባህል የበለጸገች ስትሆን የተለያዩ ብሔር ብሔረሰቦች የሚኖሩባት...
Prompt 2: አዲስ አበባ የኢትዮጵያ ዋና ከተማ (Addis Ababa the capital of Ethiopia...)	Qwen3.5: ...ነው። በዚህ አቆጣጠር በ 1979 በኢትዮጵያ እስከ 1991 .ም. ድረስ በኋላ በዝብ ከዚህ መት በኋላ... HRM-Text: ...ነው። በደቡብ በዘመነ ወይም በተሸጠቁ እንደ ዋት የዝብ ከነዚህ ዝብ በማናቸው የኤሎቹ...
Prompt 3: የሳይንስና የቴክኖሎጂ እድገት ለሀገር (Advancement of science and tech...)	Qwen3.5: ...ሲያስተዳደግ የኢኮኖሚ እና የማህበራዊ እድገት ከፍተኛ ሚና ይጫወታል። HRM-Text: ...እንዲሁም ከአረሰል አር እንደ ያላቸው ም እስከ በማስተዋ አስተኛ የአማርኛ ምሳሌ ነው።

2. Extreme Efficiency of HRM-Text HRM-Text establishes an extraordinary computational profile:

- $9.4\times$ Higher Throughput: Achieved 43,078 tokens/sec compared to 4,565 tokens/sec for the Transformer.
- $22.4\times$ Lower VRAM: Peak GPU memory allocation was only 3.43 GB, compared to 76.68 GB for Transformer.
- Competitive Convergence: Completed a full pretraining pass over 6.3M tokens in 2.5 minutes, reaching a validation perplexity of 10.83.

4.3 Qualitative Amharic Text Generation

Table 2 presents sample autoregressive text completions generated by both models on standard Amharic prompts.

Both models demonstrate strong syntactic agreement, correctly matching Amharic copular verbs (ነው, ናት, ሲሆን) with masculine and feminine subject head nouns.

5 Conclusion and Future Work

In this study, we conducted a rigorous, parameter-matched empirical comparison of 50M-class language models on the low-resource Semitic language Amharic. Our experiments demonstrate that:

1. Hybrid architectures combining L_2 -normalized linear associative memory (Gated DeltaNet) with periodic full attention achieve state-of-the-art representation quality (PPL = 9.25).
2. Dual-timescale hierarchical recurrent models (HRM-Text) provide unprecedented training throughput (43, 078 tok/s) and minimal memory footprint (3.43 GB) with minimal degradation in perplexity (PPL = 10.83).

Future research will investigate scaling these findings to 1B+ parameters and exploring cross-lingual transfer across Ge'ez-script languages including Tigrinya and Ge'ez.

References

- [1] Y. Yimam et al., “Modern NLP Benchmarks for Low-Resource Semitic Languages: The Amharic Case Study,” Transactions of the Association for Computational Linguistics (TACL), 2024.
- [2] M. L. Bender, J. D. Bowen, R. L. Cooper, and C. A. Ferguson, Language in Ethiopia, Oxford University Press, 1976.
- [3] S. M. Yimam, P. Biemann, and W. G. Weldegebriel, “Pretrained Language Models for Amharic: Analysis and Evaluation,” EACL, 2021.
- [4] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, “Attention is All You Need,” Advances in Neural Information Processing Systems (NeurIPS), 2017.
- [5] Qwen Team, “Qwen3.5 Technical Report: Next-Generation Hybrid Models and Multi-Token Architectures,” arXiv preprint, 2025.
- [6] S. Yang, B. Wang, Y. Shen, R. Panda, and Y. Kim, “Gated Delta Networks: Improving Linear Attention with Data-Dependent Decay and Recurrent Updates,” International Conference on Learning Representations (ICLR), 2024.
- [7] Y. Wang et al. (Sapient Intelligence), “HRM-Text: Hierarchical Reasoning Models for Scalable Language Pretraining,” arXiv:2605.20613, 2026.
- [8] A. Gu and T. Dao, “Mamba: Linear-Time Sequence Modeling with Selective State Spaces,” arXiv:2312.00752, 2023.
- [9] P. Joshi, S. Santy, A. Budhiraja, K. Bali, and M. Choudhury, “The State and Fate of Linguistic Diversity and Inclusion in the NLP World,” ACL, 2020.